

CATS⁺

Competishun Advance

Training Sessions

(Lecture - VII)

Ques Let $x, y, z \in \mathbb{R}$, satisfying $x + y + z = 9xyz$ also define $f(t) = \frac{1-3t^2}{1-27t^2}$, Now if $\frac{f(x)}{y \cdot z} + \frac{f(y)}{x \cdot z} + \frac{f(z)}{x \cdot y} = \lambda \cdot f(x) \cdot f(y) \cdot f(z)$ then find value of λ .

Soln.

$$\sum \frac{f(x)}{yz} = \lambda \prod f(x)$$

$$\Rightarrow \sum \frac{x f(x)}{xyz} = \lambda \prod f(x) \Rightarrow \sum x \cdot f(x) = \lambda xyz \cdot \prod f(x)$$

$$\Rightarrow \sum x f(x) = \lambda \prod (x f(x)) \quad \text{--- (1)}$$

Given

$$\begin{cases} x + y + z = 9xyz \\ \Rightarrow 3x + 3y + 3z = (3x)(3y)(3z) \end{cases}$$

$$\text{Let } 3x = \tan(\alpha)$$

$$3y = \tan(\beta)$$

$$\textcircled{2} 3z = \tan(\gamma)$$

Given relation $\Rightarrow$

$$\tan(\alpha) + \tan(\beta) + \tan(\gamma)$$

$$= \tan(\alpha) \cdot \tan(\beta) \tan(\gamma)$$

$$\Rightarrow \tan(\alpha + \beta + \gamma) = 0$$

$$\Rightarrow \alpha + \beta + \gamma = n\pi \quad \text{--- (2)}$$

$$\textcircled{1} \Rightarrow \sum x f(x) = \prod x f(x)$$

$$x \cdot f(x) = x \cdot \left(\frac{1 - 3x^2}{1 - 27x^2} \right) = \frac{x - 3x^3}{1 - 27x^2}$$

$$= \frac{1}{9} \left(\frac{3 \cdot 3x - 27x^3}{1 - 3(3x)^2} \right) = \frac{1}{3} \left(\frac{3 \cdot (3x) - (3x)^3}{1 - 3 \cdot (3x)^2} \right)$$

$$= \frac{1}{9} \left(\frac{3 \tan(\alpha) - \tan^3(\alpha)}{1 - 3 \tan^2(\alpha)} \right) = \frac{1}{9} \tan(3\alpha)$$

$$\textcircled{1} \Rightarrow \sum x f(x) = \lambda \cdot \prod (x f(x))$$

$$\sum \frac{\tan(3\alpha)}{9} = \lambda \prod \left(\frac{\tan(3\alpha)}{9} \right)$$

$$\underline{\sum \tan(3\alpha)} = \boxed{\frac{9\lambda}{9^3}} \cdot \prod \tan(3\alpha)$$

$$\frac{9\lambda}{9^3} = 1 \Rightarrow \lambda = 9^2$$

$$\Rightarrow \boxed{\lambda = 81}$$

From ②

$$\alpha + \beta + \gamma = n\pi$$

$$\Rightarrow 3\alpha + 3\beta + 3\gamma = 3n\pi$$

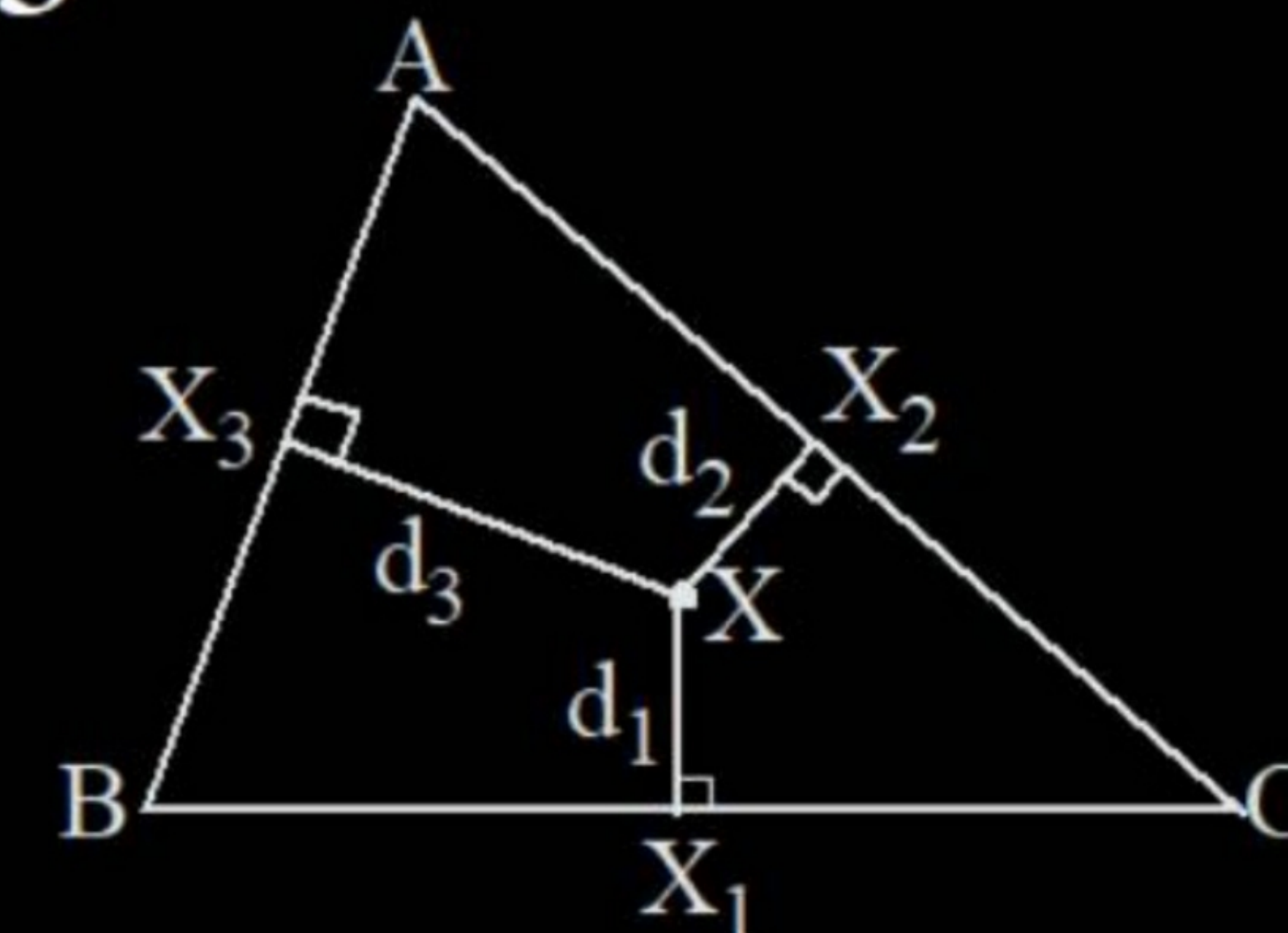
$$\Rightarrow \tan(3\alpha + 3\beta + 3\gamma) = \tan(3n\pi) = 0$$

$$\Rightarrow S_1 = S_3$$

$$\tan(3\alpha) = \prod \tan(3\alpha)$$

Paragraph for question nos. 1 to 3

In $\triangle ABC$ as shown, $XX_1 = d_1$; $XX_2 = d_2$; $XX_3 = d_3$ and X is the centre of the circumscribed circle around the $\triangle ABC$. a , b and c as usual are the sides BC , CA and AB respectively.



Q.1 If $\lambda \left(\frac{a}{d_1} + \frac{b}{d_2} + \frac{c}{d_3} \right) = \frac{abc}{d_1 d_2 d_3}$, then the value of ' λ ' is equal to

(A) 1

(B) 2

☒ (C) 4

(D) 8

Q.2 If R is the radius of the circumcircle of the $\triangle ABC$ and $\underline{a(d_2 + d_3) + b(d_3 + d_1) + c(d_1 + d_2) = kR(a + b + c)}$ then the value of ' k ' is

☒ (A) 1

(B) $\frac{1}{2}$

(C) $\frac{1}{3}$

(D) 2

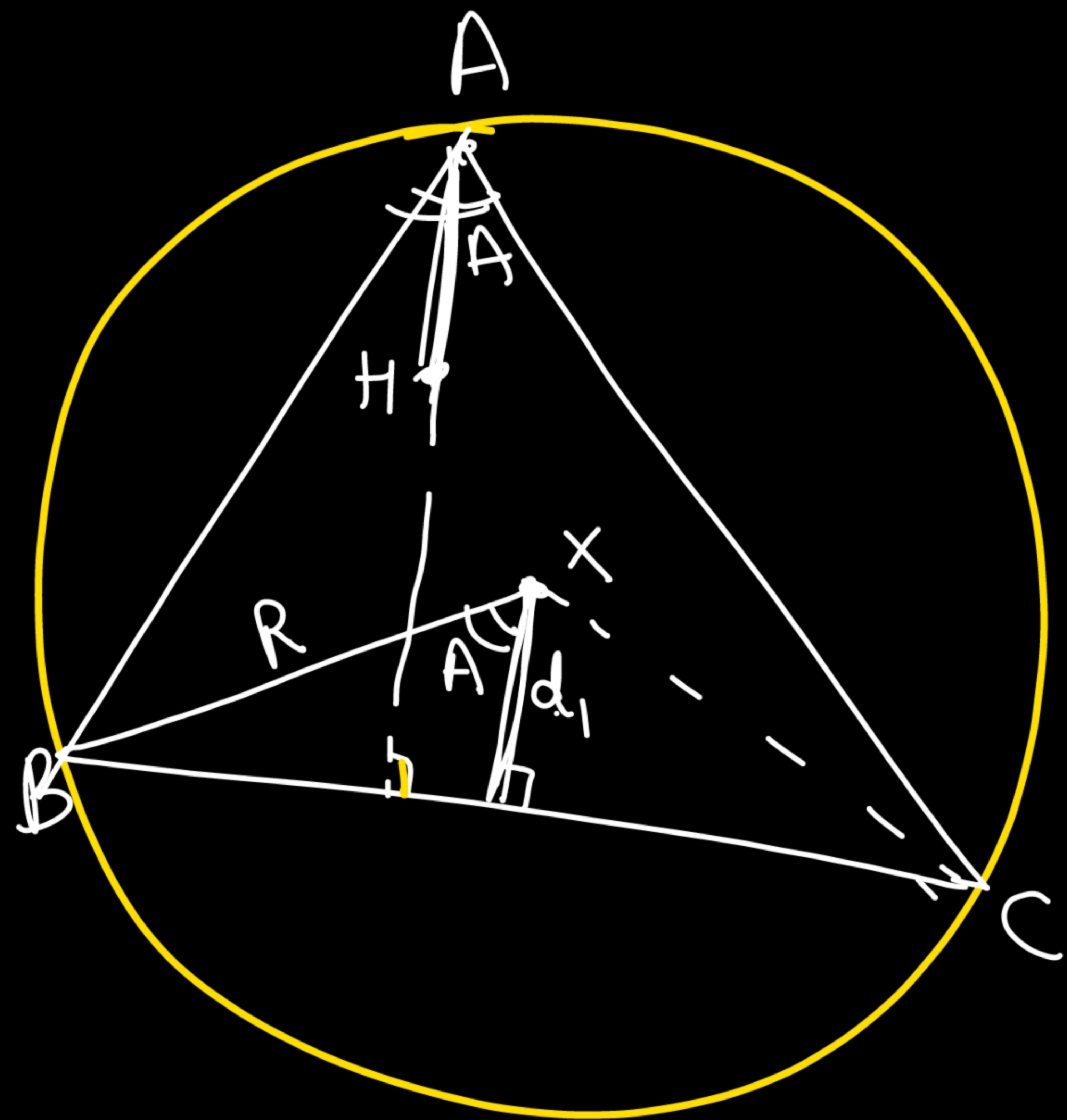
Q.3 Let h_a , h_b and h_c are the altitudes of the $\triangle ABC$ from the angular points A , B and C respectively. If $(a^2 + b^2 + c^2) = t(h_a d_1 + h_b d_2 + h_c d_3)$ then ' t ' equals

(A) 1

(B) 2

(C) 3

☒ (D) 4



$$d_1 = R \cos(A)$$

$$\textcircled{i} \quad \lambda \sum \frac{a}{d_1} = \frac{\pi a}{\pi d_1} = \pi \left(\frac{a}{d_1} \right)$$

$$\frac{a}{d_1} = \frac{2R \sin(A)}{R \cos(A)} = 2 \tan(A) \quad \Rightarrow \quad \tan(A) = \frac{a}{2d_1}$$

$$\sum \tan(A) = \pi \tan(A)$$

$$\sum \frac{a}{2d_1} = \pi \frac{a}{2d_1} \Rightarrow \frac{1}{2} \sum \frac{a}{d_1} = \frac{1}{2} \pi \frac{a}{d_1}$$

$$4 \sum \frac{a}{d_1} = \pi \frac{a}{d_1}$$

$$\lambda = 4$$

ii)

$$\sum a(d_2 + d_3) = \underline{k \cdot R \cdot (a + b + c)}$$

$$= \sum 2R \sin(A) \cdot (R \cos(B) + R \cos(C))$$

$$= 2R^2 \sum (\sin(A) \cos(B) + \sin(A) \cos(C))$$

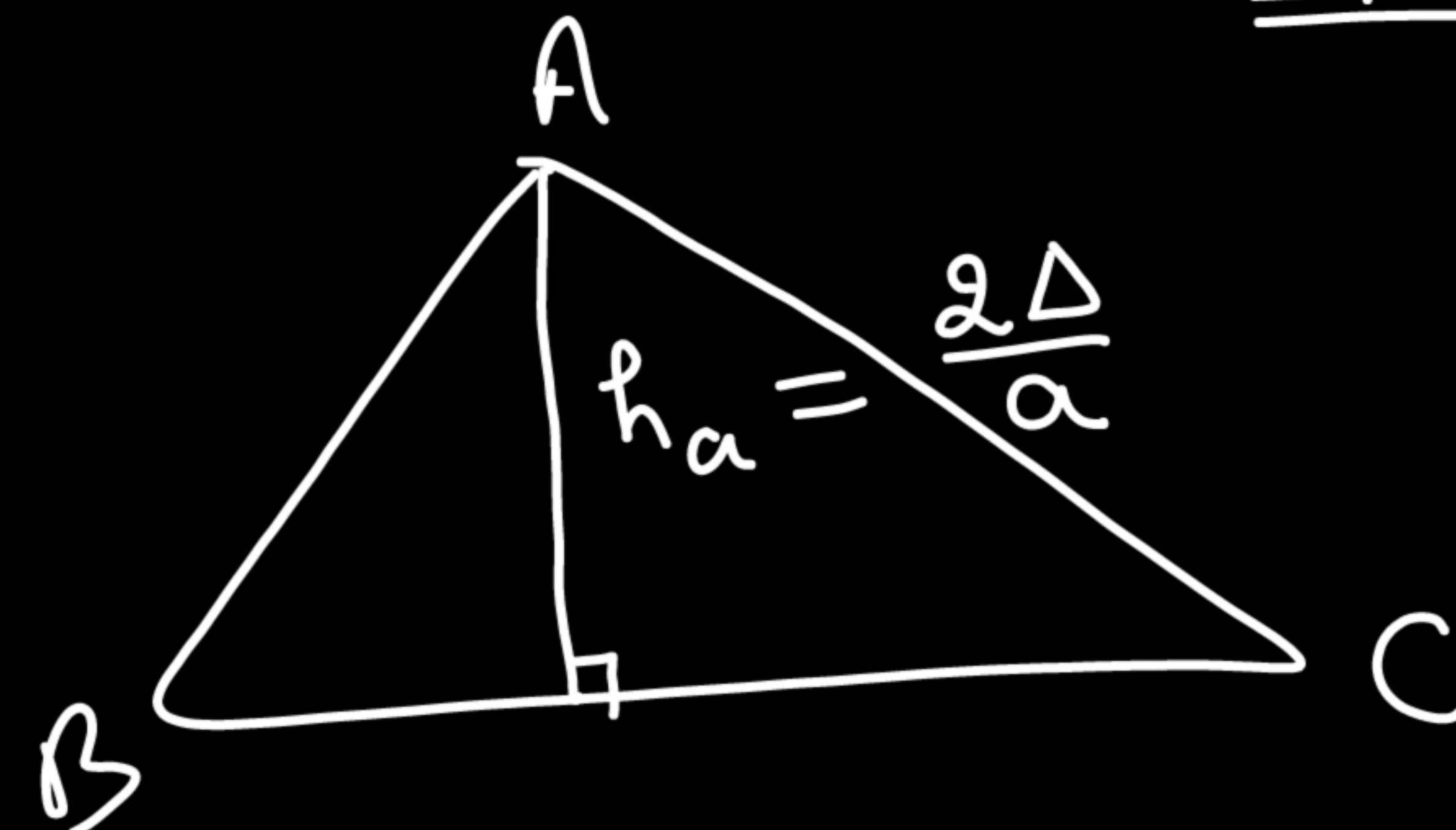
$$= 2R^2 \left(\frac{\sin(A) \cos(B) + \sin(A) \cos(C)}{+ \sin(B) \cos(A) + \sin(B) \cos(C)} \right. \\ \left. + \sin(C) \cos(A) + \sin(C) \cos(B) \right)$$

$$= 2R^2 (\sin(A+B) + \sin(B+C) + \sin(C+A)) \\ = R(a + b + c)$$

$$\boxed{K=1}$$

iii)

$$a^2 + b^2 + c^2 = t(\underline{h_a d_1} + h_b d_2 + h_c d_3)$$



$$h_a d_1 = \frac{2\Delta}{a} \cdot R \cos(A)$$

$$= \frac{2\Delta R \cos(A)}{2R \sin(A)} = \Delta \cot(A)$$

$$t(h_a \cdot d_1 + h_b \cdot d_2 + h_c \cdot d_3) = a^2 + b^2 + c^2.$$

$$= t(\sum h_a d_1)$$

$$= t(\sum \Delta \cdot \cot(A))$$

$$= \Delta \cdot t \cdot \sum \cot(A)$$

$$= \cancel{\Delta} \cdot t \cdot \left(\frac{a^2 + b^2 + c^2}{\cancel{4\Delta}} \right)$$

$$\frac{t}{4} = 1 \Rightarrow \boxed{t=4}$$

$$\sum \cot(A) = \frac{\sum a^2}{4\Delta}.$$

$$= \sum \frac{\cot(A)}{\sin(A)} = \sum \frac{b^2 + c^2 - a^2}{\underbrace{2bc \sin(A)}}_{4\Delta}$$

$$= \frac{1}{4\Delta} \sum (b^2 + c^2 - a^2)$$

$$= \frac{\sum a^2}{4\Delta}$$

$$\Delta = \frac{1}{2} bc \sin(A)$$

Que

Column-I

Column-II

(A) If the value of $(\tan 18^\circ)(\sin 36^\circ)(\cos 54^\circ)(\tan 72^\circ)(\tan 108^\circ) \times (\cos 126^\circ)(\sin 144^\circ)(\tan 162^\circ)(\cos 180^\circ)$ is $k \sin^2 18^\circ$, then 'k' has the value equal to

(B) If $\sin^3 x \cos 3x + \cos^3 x \sin 3x = \frac{3}{8}$, then the value of $\sin 4x$ is

(C) For all permissible values of x, the maximum value of the

$$f(x) = \frac{5 \sin^3 x \cos x}{\tan^2 x + 1}, \text{ is}$$

(P) $\frac{1}{2}$

(Q) $\frac{3}{4}$

(R) $\frac{5}{4}$

(S) $\frac{5}{8}$

A $\rightarrow$ R.

B $\rightarrow$ P.

C $\rightarrow$ S.

$90-36^\circ$
 $90-18^\circ$
 $(90+18^\circ)$
 $180-36^\circ$
 $90+36^\circ$
 $180-18^\circ$

~~$\tan(8^\circ) \cdot \sin(36^\circ) \cdot \sin(36^\circ) \cdot \cot(18^\circ)$~~
 ~~$(-\cot(18^\circ)) \cdot (-\sin(36^\circ)) \cdot (\sin(36^\circ)) \cdot (-\tan(18^\circ)) \cdot (-1)$~~
 $= \sin^4(36^\circ)$

$$K = \sin^2(18^\circ)$$

$$= \sin^4(36^\circ)$$

$$= \sin^2(36^\circ) \cdot \sin^2(36^\circ)$$

$$= \sin^2(36^\circ) \cdot \left(2 \sin(18^\circ) \cos(18^\circ) \right)^2$$

$$= \sin^2(36^\circ) \cdot 4 \cdot \sin^2(18^\circ) \cdot \cos^2(18^\circ)$$

$$K = 4 \sin^2(36^\circ) \cdot \cos^2(18^\circ)$$

$$= \left(1 - \cos(72^\circ) \right) \left(1 + \cos(36^\circ) \right)$$

$$= \left(1 - \frac{\sqrt{5}-1}{4} \right) \left(1 + \frac{\sqrt{5}+1}{4} \right) = \left(\frac{5-\sqrt{5}}{4} \right) \left(\frac{5+\sqrt{5}}{4} \right)$$

$$= \frac{25-5}{16} = \frac{20}{16} = \frac{5}{4}$$

$$\textcircled{B} \quad \sin^3(x) \cdot \cos(3x) + \cos^3(x) \cdot \sin(3x) = \frac{3}{8}$$

$$S^3 \cdot (\underline{4C^3 - 3C}) + C^3(3S - 4S^3) = \frac{3}{8}$$

$$\rightarrow S^3 \cdot C \cdot (4C^2 - 3) + C^3 \cdot S(3 - 4S^2)$$

$$= S \cdot C \left(S^2(4C^2 - 3) + C^2(3 - 4S^2) \right)$$

$$= \frac{\sin(2x)}{2} \left(\cancel{4S^2C^2} - \underbrace{3S^2 + 3C^2}_{=3(C^2 - S^2)} - \cancel{4C^2S^2} \right)$$

$$= 3(C^2 - S^2)$$

$$= 3\cos(2x)$$

$$= \frac{3}{2} \sin(2x) \cos(2x) = \frac{3}{4} \sin(4x)$$

$$\cancel{\frac{3}{4}} \sin(4x) = \cancel{\frac{3}{8}}$$

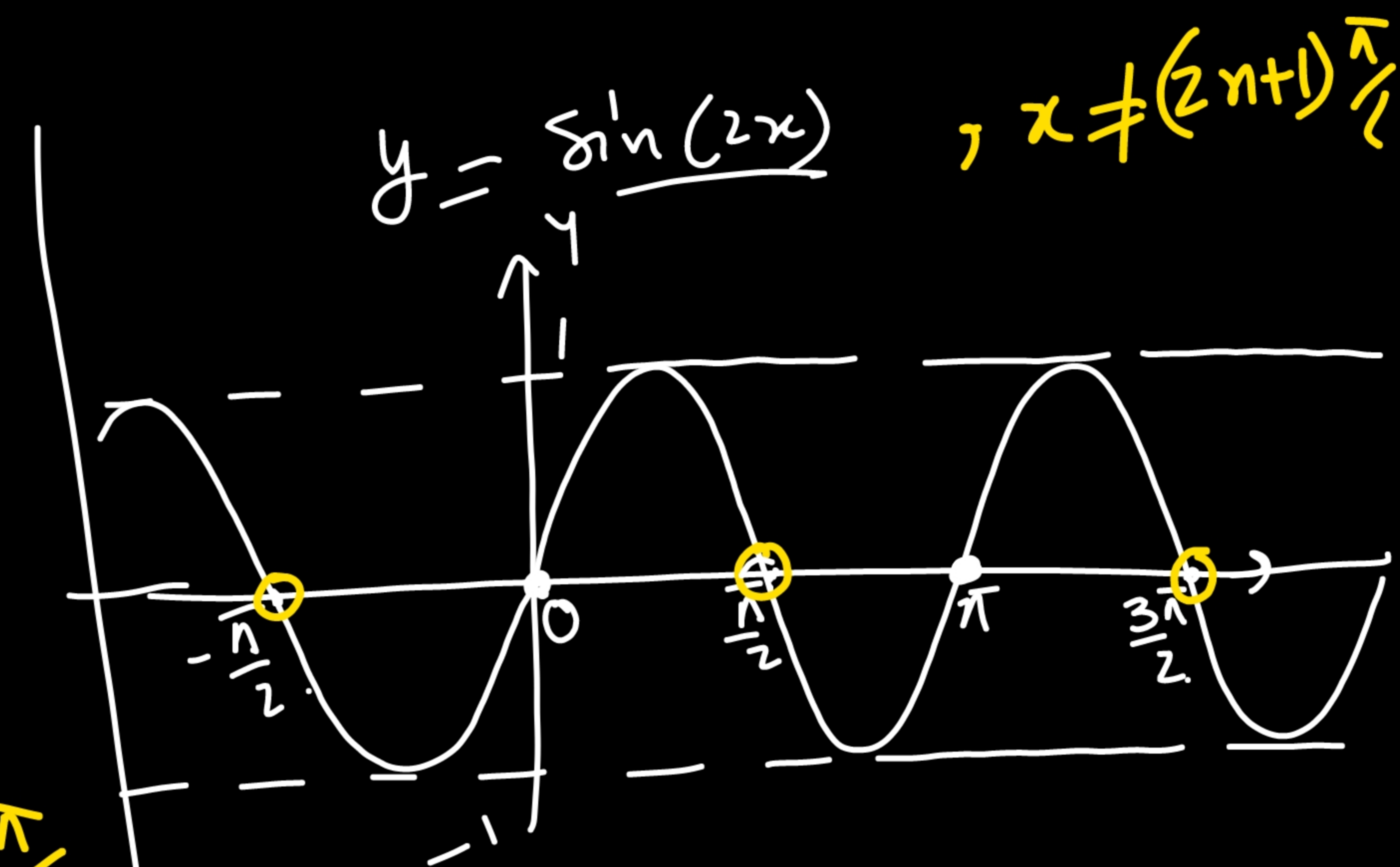
$$\sin(4x) = \frac{1}{2}$$

$$f(x) = \frac{5 \sin^3(x) \cdot \cos(x)}{1 + \tan^2(x)}$$

$$= \frac{5 \sin^3(x) \cdot \cos(x)}{\sec^2(x)}$$

$$= 5 \sin^3(x) \cdot \cos^3(x), \quad x \neq (2n+1)\frac{\pi}{2}$$

$$= \frac{5}{8} \sin^3(2x), \quad x \neq (2n+1)\frac{\pi}{2}$$



$$\sin(2x) \in [-1, 1]$$

$$\sin^3(2x) \in [-1, 1]$$

$$\frac{5}{8} \sin^3(2x) \in \left[-\frac{5}{8}, \frac{5}{8}\right]$$

Ques

A cricket player played n ($n > 1$) matches during his career and made a total of $\frac{(n^2 - 12n + 39)(4 \cdot 6^n - 5 \cdot 3^n + 1)}{5}$ runs. If T_r represent the runs made by the player in r^{th} match such

that $T_1 = 6$ and $T_r = 3T_{r-1} + 6^r$, $2 \leq r \leq n$ then find n .

Sol

Total Runs

$$= \frac{(n^2 - 12n + 39)(4 \cdot 6^n - 5 \cdot 3^n + 1)}{5}$$
$$= T_1 + T_2 + T_3 + \dots + T_n$$

$$T_r = 3T_{r-1} + 6^r$$
$$\Rightarrow T_r - 3T_{r-1} = 6^r$$
$$\Rightarrow \frac{T_r}{3^r} - \frac{T_{r-1}}{3^{r-1}} = 2^r$$

$r=2, 3, \dots, n$

$$\frac{T_2}{3^2} - \frac{T_1}{3} = 2^2$$
$$\frac{T_3}{3^3} - \frac{T_2}{3^2} = 2^3$$
$$\vdots$$

$$\begin{aligned}
\sum_{r=1}^n T_r &= \sum_{r=1}^n 2 \cdot (6^r - 3^r) \\
&= 2 \sum_{r=1}^n (6^r - 3^r) \\
&= 2 \left(6 \cdot \frac{(6^n - 1)}{6 - 1} - \frac{3 \cdot (3^n - 1)}{3 - 1} \right) \\
&= 2 \left(\frac{6(6^n - 1) - 3 \cdot (3^n - 1)}{2} \right) \\
&= \cancel{6}^3 \left(\frac{4(6^n - 1) - 5(3^n - 1)}{5 \times 2} \right) \\
&= \frac{3}{5} (4 \cdot 6^n - 5 \cdot 3^n + 1)
\end{aligned}$$

$$\frac{T_n}{3^n} - \frac{T_1}{3} = 2^2 + 2^3 + \dots + 2^n$$

$$\frac{T_n}{3^n} - 2 = 2^2 + 2^3 + \dots + 2^n$$

$$T_n = 3^n (2 + 2^2 + 2^3 + \dots + 2^n)$$

$$= 3^n \left(2 \cdot \left(\frac{2^n - 1}{2 - 1} \right) \right)$$

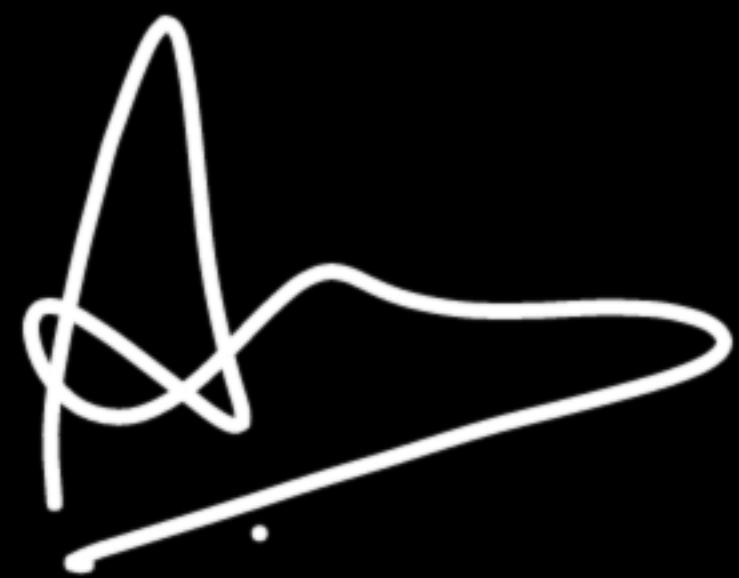
$$T_n = 2(6^n - 3^n)$$

$$n^2 - 12n + 39 = 3$$

$$n^2 - 12n + 36 = 0$$

$$\hookrightarrow (n-6)^2 = 0.$$

$$\boxed{n=6}$$



One

If p, q, r each are positive rational number such that $p > q > r$ and the quadratic equation $(p + q - 2r)x^2 + (q + r - 2p)x + (r + p - 2q) = 0$ has a root in $(-1, 0)$ then which of the following statement(s) holds good?

✓ (A) $\frac{r+p}{q} < 2 \Rightarrow \underline{r+p} < \underline{2q}$

✓ (B) Both roots of the given quadratic equation are rational.

✓ (C) The equation $px^2 + 2qx + r = 0$ has real and distinct roots.

(D) The equation $px^2 + 2qx + r = 0$ has no real roots.

$\Rightarrow \alpha \in (-1, 0)$

$p > q > r > 0$
rational nos.

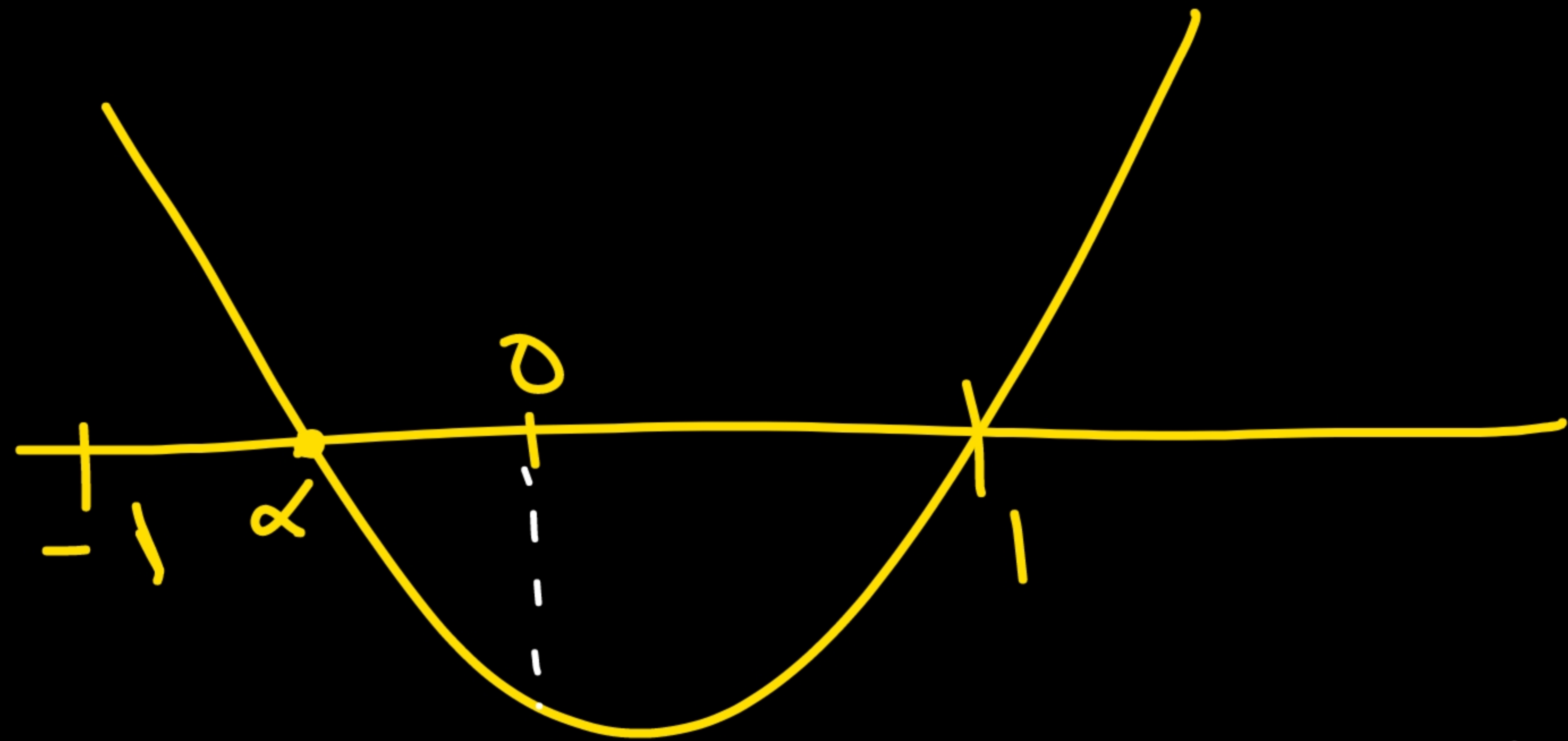
$(p+q-2r)x^2 + (q+r-2p)x + (r+p-2q) = 0$

$\underbrace{x=1}_{\text{cancel}} \rightarrow p+q-2r + q+r-2p + r+p-2q = 0$
 $0 = 0$

$\Rightarrow 1, \alpha$

$p = 1 \cdot \alpha = \frac{r+p-2q}{p+q-2r}$

as $p, q > r \Rightarrow p+q-2r > 0 \Rightarrow$ Parabola Upward



as it is parabola Upwards

⑧ '0' lies b/w the roots

hence $f(0) < 0$.

$$r+p-2q < 0 \Rightarrow 2q > r+p.$$

$$\frac{r+p}{q} < 2$$

$$px^2 + 2qx + r = 0.$$

$$D_1 = 4q^2 - 4pr$$

$$= 4(q^2 - pr)$$

$$= (2q)^2 - 4pr.$$

$$> (r+p)^2 - 4pr.$$

$$= (r-p)^2 > 0.$$

$$\boxed{D_1 > 0}$$

Ques Let $f_n(\theta) = \sum_{n=0}^n \frac{1}{4^n} \sin^4(2^n \theta)$. Then which of the following

alternative(s) is/are correct?

$$\frac{\sqrt{2}-1}{2\sqrt{2}}$$

~~(A)~~ $f_2\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

~~(B)~~ $f_3\left(\frac{\pi}{8}\right) = \frac{2+\sqrt{2}}{4}$

(C) $f_4\left(\frac{3\pi}{2}\right) = 1$

(D) $f_5(\pi) = 0$

Sol
$$f_n(\theta) = \sum_{n=0}^n \frac{1}{4^n} \sin^4(2^n \theta)$$

$$f_n(\theta) = \sum_{n=0}^{\infty} \frac{\sin^4(2^n \theta)}{4^n} = \sum_{n=0}^{\infty} \frac{\sin^2(2^n \theta)}{4^n} \sin^2(2^n \theta).$$

$$= \sum_{n=0}^{\infty} \frac{\sin^2(2^n \theta) (1 - \cos^2(2^n \theta))}{4^n}$$

$$= \sum_{n=0}^{\infty} \frac{\sin^2(2^n \theta) - \sin^2(2^n \theta) \cdot \cos^2(2^n \theta)}{4^n}$$

$$= \sum_{n=0}^{\infty} \left(\underbrace{\frac{\sin^2(2^n \theta)}{4^n}}_{V_n} - \underbrace{\frac{\sin^2(2^{n+1} \theta)}{4^{n+1}}}_{V_{n+1}} \right)$$

$$\begin{aligned} & \left[\frac{\sin^2(\alpha) \cdot \cos^2(\alpha)}{4} \right] \\ & \quad \downarrow \alpha = 2^n \theta \\ & \sin^2(2^n \theta) \cdot \cos^2(2^n \theta) \\ & \quad \downarrow \\ & \quad \underline{\underline{\frac{\sin^2(2 \cdot 2^n \theta)}{4}}} \end{aligned}$$

$$f_n(\theta) = \sin^2(\theta) - \frac{\sin^2(2^{n+1}\theta)}{4^{n+1}}$$

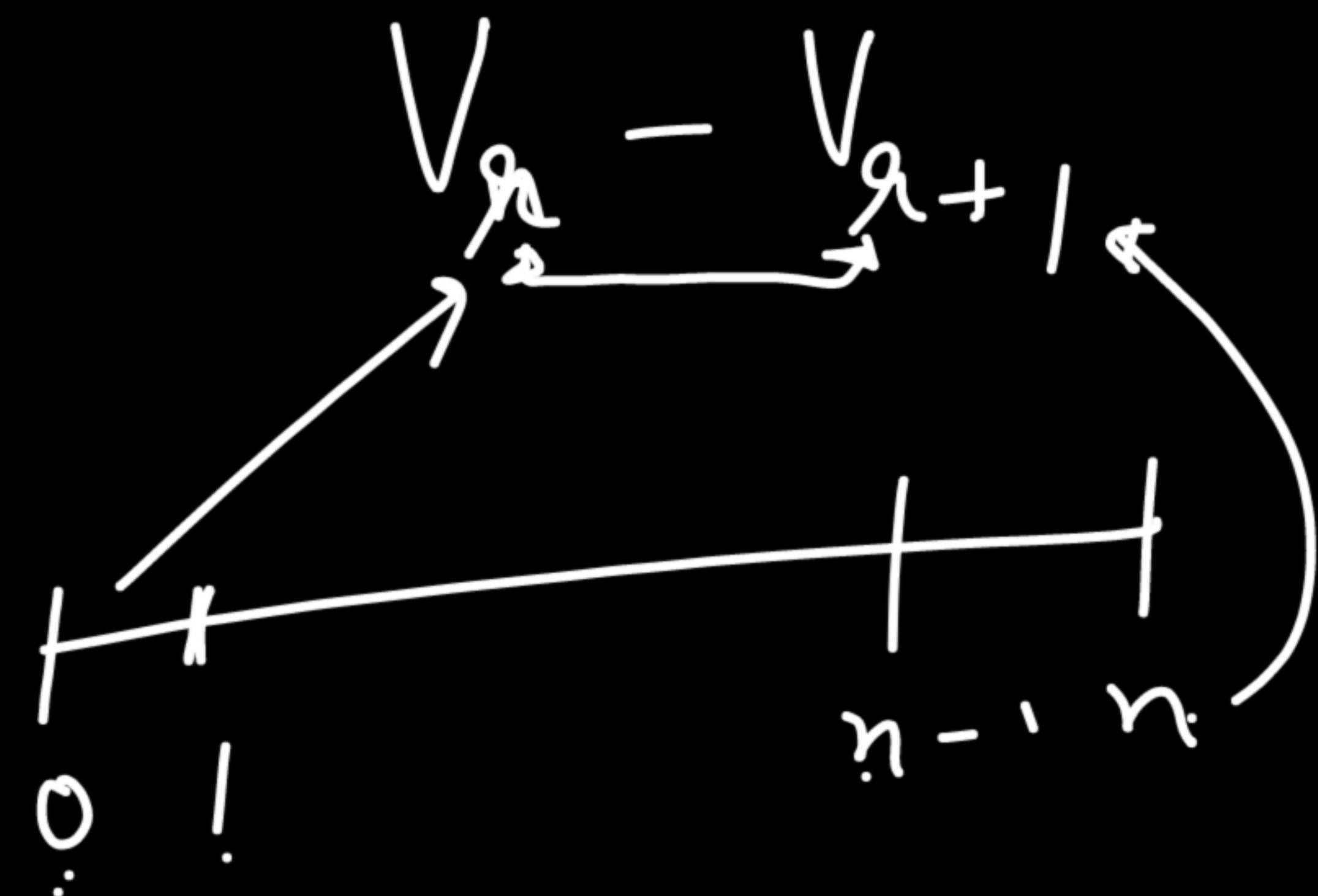
$$f_2\left(\frac{\pi}{4}\right) = \sin^2\left(\frac{\pi}{4}\right) - \frac{\sin^2\left(8 \cdot \frac{\pi}{4}\right)}{4^3}$$

$$= \frac{1}{2} - 0 = \frac{1}{2}$$

$$f_3\left(\frac{\pi}{8}\right) = \sin^2\left(\frac{\pi}{8}\right) - \frac{\sin^2\left(16 \cdot \frac{\pi}{8}\right)}{4^4}$$

$$= \sin^2\left(\frac{\pi}{8}\right) = \frac{1 - \cos\left(\frac{\pi}{4}\right)}{2} = \frac{1 - \frac{1}{\sqrt{2}}}{2}$$

$$= \frac{\sqrt{2} - 1}{2\sqrt{2}}$$



$$\sum_{r=0}^n (V_r - V_{r+2})$$

$$= V_0 + V_1 - V_{n+2} - V_{n+1}$$

$$f_n(0) = \sin^2(0) - \frac{\sin^2(2^{n+1} \cdot 0)}{4^{n+1}}$$

$$f_4\left(\frac{3\pi}{2}\right) = 1 - \frac{\sin^2\left(32 \cdot \frac{3\pi}{2}\right)}{4^5}$$

= 1

0

$$f_5(\pi) = 0 - 0 = 0.$$